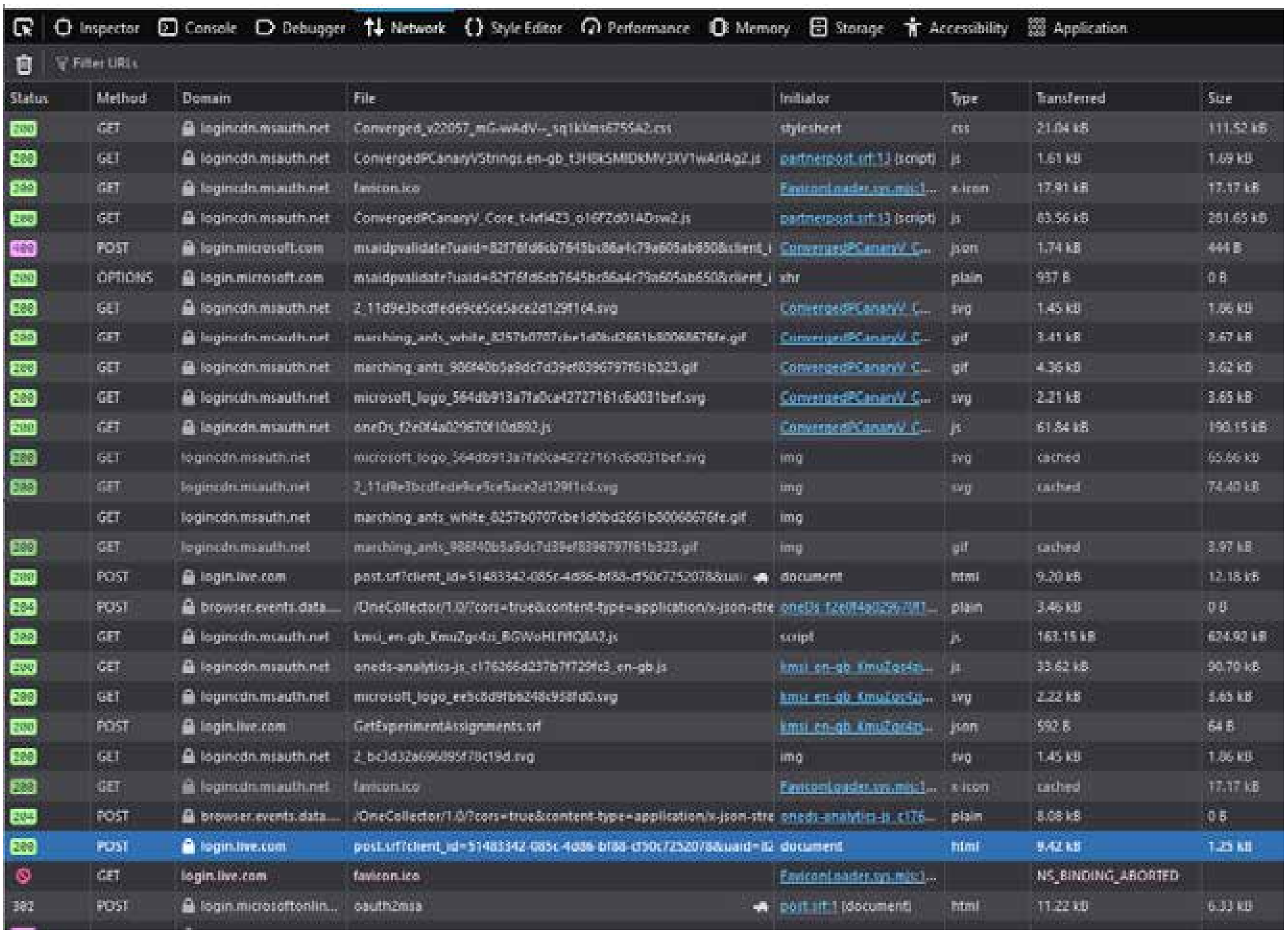

Inspector | Console | Debugger | Network | Style Editor | Performance | Memory | Storage | Accessibility | Application

Filter URLs

| Status | Method | Domain | File | Initiator | Type | Transferred | Size |
|---|---|---|---|---|---|---|---|
| 200 | GET | logincdn.msauth.net | Converged_v22057_mG-wAdV--_sq1kXms6755A2.css | stylesheet | css | 21.04 kB | 111.52 kB |
| 200 | GET | logincdn.msauth.net | ConvergedPCanaryVStrings.en-gb_t3H8kSMlDkMV3XV1wAriAg2.js | partnerpost.srf:13 (script) | js | 1.61 kB | 1.69 kB |
| 200 | GET | logincdn.msauth.net | favicon.ico | FaviconLoader.sys.mjs:1... | x-icon | 17.91 kB | 17.17 kB |
| 200 | GET | logincdn.msauth.net | ConvergedPCanaryV_Core_t-lvfl423_o16fZd01ADsw2.js | partnerpost.srf:13 (script) | js | 83.56 kB | 281.65 kB |
| 400 | POST | login.microsoft.com | msaidpvalidate?uaid=82f76fd6cb7645bc86a4c79a605ab650&client_i | ConvergedPCanaryV_C... | json | 1.74 kB | 444 B |
| 200 | OPTIONS | login.microsoft.com | msaidpvalidate?uaid=82f76fd6cb7645bc86a4c79a605ab650&client_i | xhr | plain | 937 B | 0 B |
| 200 | GET | logincdn.msauth.net | 2_11d9e3bcdfede9ce5ce5ace2d129f1c4.svg | ConvergedPCanaryV_C... | svg | 1.45 kB | 1.06 kB |
| 200 | GET | logincdn.msauth.net | marching_ants_white_8257b0707cbe1d0bd2661b80068676fe.gif | ConvergedPCanaryV_C... | gif | 3.41 kB | 2.67 kB |
| 200 | GET | logincdn.msauth.net | marching_ants_986f40b5a9dc7d39ef8396797f61b323.gif | ConvergedPCanaryV_C... | gif | 4.36 kB | 3.62 kB |
| 200 | GET | logincdn.msauth.net | microsoft_logo_564db913a7fa0ca42727161c6d031bef.svg | ConvergedPCanaryV_C... | svg | 2.21 kB | 3.65 kB |
| 200 | GET | logincdn.msauth.net | oneDs_f2e0f4a029670f10d892.js | ConvergedPCanaryV_C... | js | 61.84 kB | 190.15 kB |
| 200 | GET | logincdn.msauth.net | microsoft_logo_564db913a7fa0ca42727161c6d031bef.svg | img | svg | cached | 65.66 kB |
| 200 | GET | logincdn.msauth.net | 2_11d9e3bcdfede9ce5ce5ace2d129f1c4.svg | img | svg | cached | 74.40 kB |
| | GET | logincdn.msauth.net | marching_ants_white_8257b0707cbe1d0bd2661b80068676fe.gif | img | | | |
| 200 | GET | logincdn.msauth.net | marching_ants_986f40b5a9dc7d39ef8396797f61b323.gif | img | gif | cached | 3.97 kB |
| 200 | POST | login.live.com | post.srf?client_id=51483342-085c-4d86-bf88-cf50c7252078&uai | document | html | 9.20 kB | 12.18 kB |
| 204 | POST | browser.events.data.... | /OneCollector/1.0/?cors=true&content-type=application/x-json-stre | oneDs_f2e0f4a029670f1... | plain | 3.46 kB | 0 B |
| 200 | GET | logincdn.msauth.net | kmsi_en-gb_KmuZgc4zi_BGWoHLfYfQ8A2.js | script | js | 163.15 kB | 624.92 kB |
| 200 | GET | logincdn.msauth.net | oneds-analytics-js_c176266d237b7f729fc3_en-gb.js | kmsi_en-gb_KmuZgc4zi... | js | 33.62 kB | 90.70 kB |
| 200 | GET | logincdn.msauth.net | microsoft_logo_ee5c8d9fb6248c958fd0.svg | kmsi_en-gb_KmuZgc4zi... | svg | 2.22 kB | 3.65 kB |
| 200 | POST | login.live.com | GetExperimentAssignments.srf | kmsi_en-gb_KmuZgc4zi... | json | 592 B | 64 B |
| 200 | GET | logincdn.msauth.net | 2_bc3d32a696095f78c19d.svg | img | svg | 1.45 kB | 1.06 kB |
| 200 | GET | logincdn.msauth.net | favicon.ico | FaviconLoader.sys.mjs:1... | x-icon | cached | 17.17 kB |
| 204 | POST | browser.events.data.... | /OneCollector/1.0/?cors=true&content-type=application/x-json-stre | oneds-analytics-js_c176... | plain | 8.08 kB | 0 B |
| 200 | POST | login.live.com | post.srf?client_id=51483342-085c-4d86-bf88-cf50c7252078&uaid=82 | document | html | 9.42 kB | 1.25 kB |
| | GET | login.live.com | favicon.ico | FaviconLoader.sys.mjs:1... | | NS_BINDING_ABORTED | |
| 302 | POST | login.microsoftonlin... | oauth2msa | post.srf:1 (document) | html | 11.22 kB | 6.33 kB |

Figure 4: Post.srf response where payload of Oauth2msa is present

The authorisation code is returned as an HTML value, which can be found in the response body. By analysing these series of request-response modules, you can extract the necessary parameters from the response headers or payload. These parameters can then be implemented in Python code to perform backend-based sign-in to any portal, such as Microsoft Entra, without relying on a web browser.

## Authentication in low-end IoT devices

This method can be particularly useful for hardware-based IoT solutions, such as Arduino or Raspberry Pi (RPI), where authentication using FIDO2 technologies can be implemented on low-end devices. Here's a summary of how this process works.

***Authorisation code in HTML:*** The authorisation code returned in the HTML value of the response body is critical for authentication.

***Series of request-response modules:*** By closely examining the requests and responses, you can identify and extract key parameters required for the authentication process.

***Python implementation:*** Using Python, these parameters can be programmatically handled to automate the backend sign-in process, bypassing the need for a web browser.

***Application in IoT devices:*** This approach can be applied to low-end IoT devices, like Arduino or Raspberry Pi, allowing them to authenticate using FIDO2 technologies without a browser dependency.

By leveraging these techniques, developers can create more efficient and secure authentication processes for various applications, including IoT devices. This ensures robust security and streamlined operations, even on low-power hardware platforms. END


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